

Assignment 2 - Solutions

00103335: Deep Learning and Reinforcement Learning
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Problem 1. Equivalence of convexity conditions

Throughout this problem, write $d = y - x$.

(i) $\Rightarrow$ (ii). For $t \in (0, 1)$, convexity at $x + td = (1 - t)x + ty$ gives

$$f(x + td) \leq (1 - t)f(x) + tf(y).$$

Consequently,

$$\frac{f(x + td) - f(x)}{t} \leq f(y) - f(x).$$

Letting $t \downarrow 0$ and using differentiability at x yields $\nabla f(x)^\top d \leq f(y) - f(x)$, which is (ii).

(ii) $\Rightarrow$ (iii). Apply (ii) first to (x, y) and then to (y, x) :

$$f(y) - f(x) \geq \nabla f(x)^\top d, \quad f(x) - f(y) \geq -\nabla f(y)^\top d.$$

Adding gives $(\nabla f(y) - \nabla f(x))^\top d \geq 0$.

(iii) $\Rightarrow$ (ii). Let $\phi(t) = f(x + td)$ for $0 \leq t \leq 1$. For $0 \leq s < t \leq 1$, condition (iii), applied at $x + sd$ and $x + td$, gives

$$(t - s)(\phi'(t) - \phi'(s)) = (\nabla f(x + td) - \nabla f(x + sd))^\top ((t - s)d) \geq 0.$$

Thus ϕ' is nondecreasing. By the mean value theorem, $\phi(1) - \phi(0) = \phi'(c) \geq \phi'(0) = \nabla f(x)^\top d$ for some $c \in (0, 1)$. This is (ii).

(ii) $\Rightarrow$ (i). Set $z = \lambda x + (1 - \lambda)y$. By (ii), $f(x) \geq f(z) + \nabla f(z)^\top (x - z)$ and $f(y) \geq f(z) + \nabla f(z)^\top (y - z)$. Multiply these inequalities by λ and $1 - \lambda$, respectively, and add. Since $\lambda(x - z) + (1 - \lambda)(y - z) = 0$,

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(z).$$

This proves all three conditions are equivalent.

Problem 2. Equivalence of strong convexity conditions

Define $g(u) = f(u) - \frac{\mu}{2}\|u\|^2$. Then $\nabla g(u) = \nabla f(u) - \mu u$. Problem 1 applies to the differentiable function g .

(i) $\Leftrightarrow$ (ii). For $z = \alpha x + (1 - \alpha)y$, the identity

$$\|z\|^2 = \alpha\|x\|^2 + (1 - \alpha)\|y\|^2 - \alpha(1 - \alpha)\|y - x\|^2$$

shows that $g(z) \leq \alpha g(x) + (1 - \alpha)g(y)$ is exactly

$$f(z) \leq \alpha f(x) + (1 - \alpha)f(y) - \frac{\mu}{2}\alpha(1 - \alpha)\|y - x\|^2.$$

(i) $\Leftrightarrow$ (iii). The first-order convexity condition for g is

$$f(y) - \frac{\mu}{2}\|y\|^2 \geq f(x) - \frac{\mu}{2}\|x\|^2 + (\nabla f(x) - \mu x)^\top (y - x).$$

Rearranging and using $\|y\|^2 - \|x\|^2 - 2x^\top(y - x) = \|y - x\|^2$ gives precisely

$$f(y) \geq f(x) + \nabla f(x)^\top(y - x) + \frac{\mu}{2}\|y - x\|^2.$$

(i) $\Leftrightarrow$ (iv). The gradient-monotonicity condition for g is

$$(\nabla g(y) - \nabla g(x))^\top(y - x) = (\nabla f(y) - \nabla f(x))^\top(y - x) - \mu\|y - x\|^2 \geq 0,$$

which is (iv). Thus all four conditions are equivalent.

Problem 3. Further consequences of strong convexity

Let $d = y - x$ and $h = \nabla f(y) - \nabla f(x)$. Problem 2(iv) gives

$$h^\top d \geq \mu\|d\|^2. \tag{1}$$

(ii). Cauchy–Schwarz and (1) imply $\mu\|d\|^2 \leq h^\top d \leq \|h\|\|d\|$. If $d = 0$, the claim is immediate; otherwise divide by $\|d\|$ to obtain $\|h\| \geq \mu\|d\|$.

(iii). Apply the strong convexity lower bound, Problem 2(iii), with x and y interchanged:

$$f(x) \geq f(y) - \nabla f(y)^\top d + \frac{\mu}{2}\|d\|^2.$$

Therefore

$$\begin{aligned} f(y) - f(x) - \nabla f(x)^\top d &\leq h^\top d - \frac{\mu}{2}\|d\|^2 \\ &= \frac{\|h\|^2}{2\mu} - \frac{\mu}{2}\|d - \frac{h}{\mu}\|^2 \\ &\leq \frac{\|h\|^2}{2\mu}. \end{aligned}$$

This is the required upper bound.

(iv). From (ii), $\|d\| \leq \|h\|/\mu$. Thus

$$h^\top d \leq \|h\|\|d\| \leq \frac{1}{\mu}\|h\|^2.$$

No Lipschitz continuity of ∇f is needed for these conclusions.

Problem 4. Gradient descent with a decreasing step size

Write $g_t = \nabla f(x_t)$, $d_t = \|x_t - x^\star\|$, and $\Delta_t = f(x_t) - f(x^\star)$.

(a). Apply Problem 2(iii) at $(x_t, x^\star)$:

$$f(x^\star) \geq f(x_t) + g_t^\top (x^\star - x_t) + \frac{\mu}{2} d_t^2.$$

Rearranging gives $g_t^\top (x_t - x^\star) \geq \Delta_t + \frac{\mu}{2} d_t^2$.

(b). Expanding the squared distance after one update and then using (a) and $\|g_t\| \leq L$,

$$\begin{aligned} d_{t+1}^2 &= \|x_t - x^\star - \eta_t g_t\|^2 \\ &= d_t^2 - 2\eta_t g_t^\top (x_t - x^\star) + \eta_t^2 \|g_t\|^2 \\ &\leq (1 - \mu\eta_t) d_t^2 - 2\eta_t \Delta_t + \eta_t^2 L^2. \end{aligned}$$

(c). Since $\eta_t = 2/[\mu(t+1)]$, rearranging (b) gives

$$\Delta_t \leq \frac{\mu}{4} ((t-1)d_t^2 - (t+1)d_{t+1}^2) + \frac{L^2}{\mu(t+1)}.$$

Multiply by t and sum. The distance terms telescope: the positive term $t(t-1)d_t^2$ cancels the negative term from iteration $t-1$, and the initial coefficient $1(1-1)$ is zero. Consequently,

$$\begin{aligned} \sum_{t=1}^T t\Delta_t &\leq -\frac{\mu}{4} T(T+1)d_{T+1}^2 + \frac{L^2}{\mu} \sum_{t=1}^T \frac{t}{t+1} \\ &\leq \frac{TL^2}{\mu}. \end{aligned}$$

(d). The weights $w_t = 2t/[T(T+1)]$ are nonnegative and sum to one. Strong convexity implies convexity, so Jensen's inequality and (c) yield

$$\begin{aligned} f(\bar{x}_T) - f(x^\star) &\leq \sum_{t=1}^T w_t (f(x_t) - f(x^\star)) \\ &= \frac{2}{T(T+1)} \sum_{t=1}^T t\Delta_t \\ &\leq \frac{2L^2}{\mu(T+1)} = O\left(\frac{1}{T}\right). \end{aligned}$$

The argument uses only the stated bound on the gradients along the iterates; it does not assume smoothness of f .